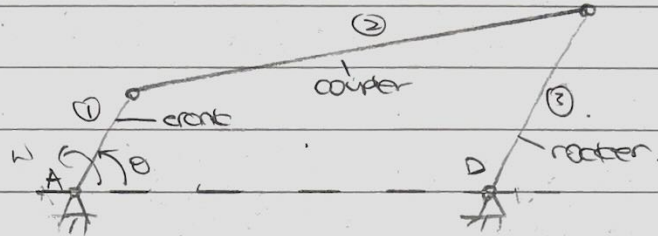


Plane mechanisms

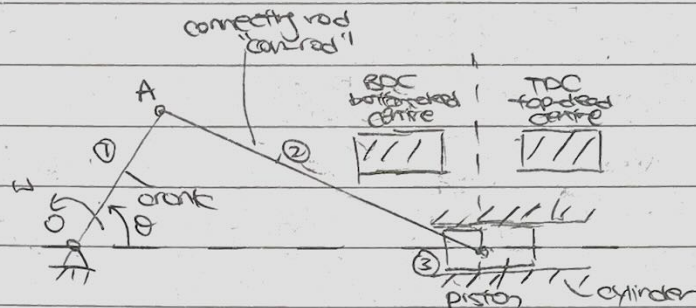
Types of plane mechanisms

- 3 common types of planar mechanisms are as follows:

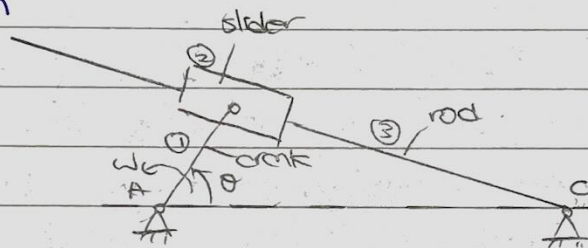
① Crank rocker.



② slider crank



③ Quick return



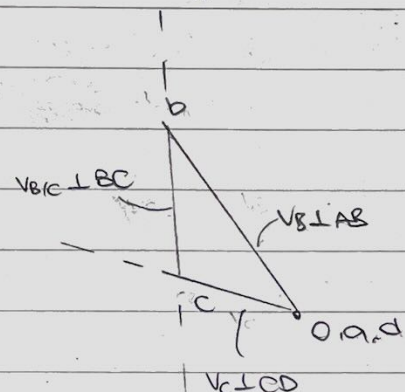
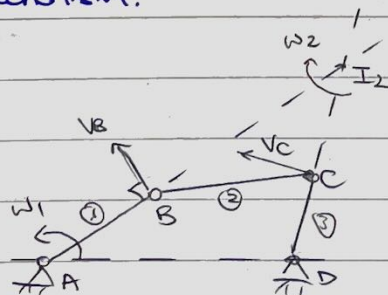
Instantaneous centres (IC) and velocity diagrams

- We can use the method of instantaneous centres to determine the velocities and angular velocities of components of a mechanism.

- If the direction of the velocities of 2 pts in a rigid body are known at a particular instant, then the IC is obt. which the body is instantaneously rotating is found at the intersection of the perpendiculars to the velocities.

- After finding the velocities, we can use a velocity diagram to graphically represent the vectors showing the rel. velocities b/w the components of the mechanism.

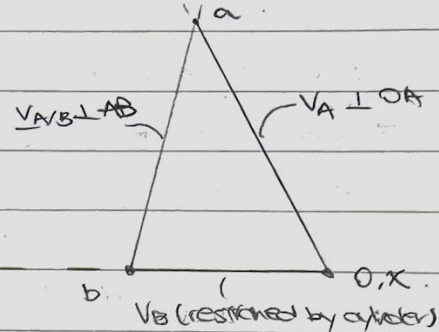
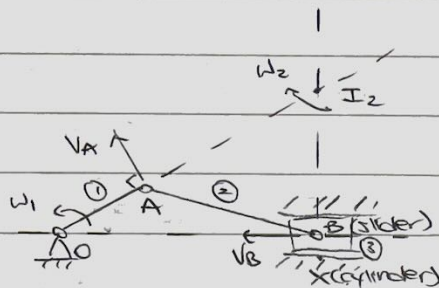
① Crank mechanism.



$$V_B = AB \cdot \omega_1 = I_2 B \cdot \omega_2 \rightarrow \omega_2 = \frac{AB}{I_2 B} \omega_1$$

$$V_C = I_2 C \omega_2 = CD \omega_3 \rightarrow \omega_3 = \frac{AB}{CD} \cdot \frac{I_2 C}{I_2 B} \omega_1$$

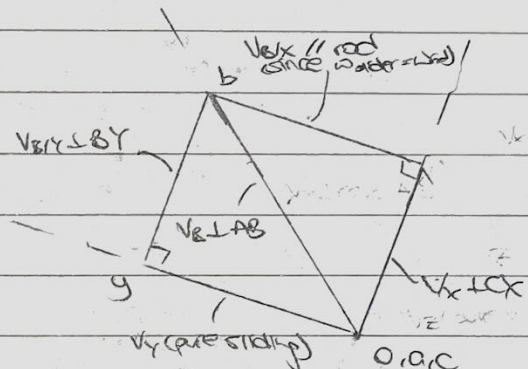
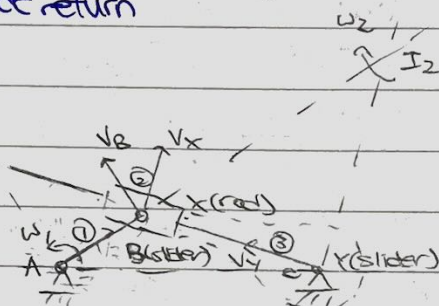
② slider crank



$$v_A = OA\omega_1 = I_2 A \omega_2 \rightarrow \omega_2 = \frac{OA}{I_2 A} \omega_1$$

$$v_B = I_2 B \omega_2 = OA \frac{I_2 B}{I_2 A} \omega_1$$

③ quick return



$$v_B = AB\omega_1 = I_2 B \cdot \omega_2 \rightarrow \omega_2 = \frac{AB}{I_2 B} \omega_1$$

* It is often helpful to extend the slider s.t. part of it is over a fixed pivot. The direction of the pt. on the slider instantaneously over the fixed pivot is known ($\omega_{rod} = \omega_{slider}$, so there must be pure sliding).

* ω_{rod} must equal ω_{slider} or else the slider wouldn't be on the rod anymore.

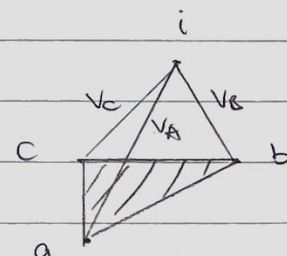
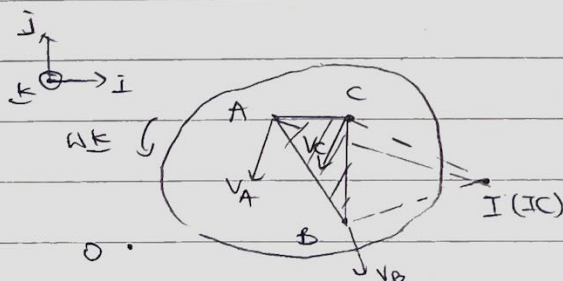
The velocity image.

- Once we have a velocity diagram, then immediate pts on a rigid lamina that has angular velocity ω can be obtained by inspection using the velocity image.

- Suppose we know the velocities of pts A, B on the lamina, and they are placed on the velocity diagram. We can locate the velocity of pt. C on the diagram by noting that $\Delta ABC \sim \Delta abc$. (Δabc is rotated 90° from ΔABC in the dir. of ω).

$$\vec{v}_B = \vec{v}_A + \vec{r}_{BA} \times \omega \rightarrow \vec{v}_B = \vec{v}_A + \vec{r}_{BA} \times \omega = \vec{v}_A + \omega \times \vec{r}_{BA}$$

$$\vec{v}_C = \vec{v}_A + \vec{r}_{CA} \times \omega \rightarrow \vec{v}_C = \vec{v}_A + \vec{r}_{CA} \times \omega = \vec{v}_A + \omega \times \vec{r}_{CA}$$



* The image tri. is useful for finding velocities from edges / within areas of a velocity diagram

Virtual power

- Mechanisms are used to transmit force and power $\rightarrow$ we want to calculate the torque req. to drive a mechanism, or calculate the effect of friction of sliders and joints.
- First consider a system of parts moving slowly w/o friction $\rightarrow$ doesn't absorb any energy and KE can be ignored. The net work done on the mechanism as it moves by a small amt is zero.
- Now consider all forces F_i and torques T_i (inc. friction) acting on the mechanism. In a small time interval δt , there are resulting displacements δx_i and rotations $\delta \theta_i$.
- Over the time interval δt , the net work done on the mechanism is zero.

$$\sum_i F_i \cdot \delta x_i + \sum_i T_i \delta \theta_i = 0 \quad \leftarrow \text{for planar motion, } T_i \text{ and } \delta \theta_i \text{ are in the } \perp \text{ direction,}$$

Dividing by δt , and noting that $\frac{\delta x_i}{\delta t} = v_i$, $\frac{\delta \theta_i}{\delta t} = \omega_i$,

$$\sum_i F_i \cdot v_i + \sum_i T_i \omega_i = 0$$

- We can use IC/velocity diagrams to find velocities v_i and angular velocities ω_i of the pt. of application of all forces and torques.

- The following pts are important when solving virtual power problems:

$\rightarrow$ Reaction forces of fixed pivots do no work as they don't have displacements.

$\rightarrow$ Friction of sliders are ext. forces $\rightarrow$ use rel. sliding velocity.

$\rightarrow$ Friction of pivots are ext. torques $\rightarrow$ use rel. angular velocity.

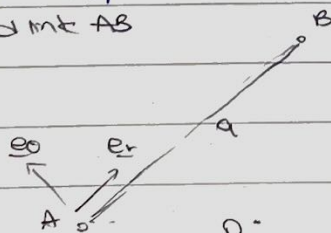
} friction always opposes the motion, i.e. -ve term

* It is helpful to tabulate all the forces / torque (esp. for friction problems).

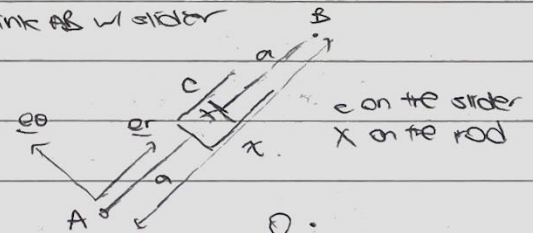
Acceleration diagrams and acceleration image.

- After finding the accelerations, we can use an acceleration diagram to graphically represent the vectors showing the rel. accelerations b/w the components of the mechanism.
- Any rigid body in the form of a polygon ABC will have a similar polygon abc in the acceleration diagram (but rotated at an angle that is not necessarily 90°).
- In mechanisms, the rel. motion b/w two pts is well defined. Two common cases are:

rigid link AB



link AB w/ slider



- Generally, we will use the expression for the rel. acceleration b/w two pts. The rel. acceleration of pt 2 rel. to pt 1 is

$$\ddot{r}_{2/1} = (\ddot{r} - r\ddot{\theta}^2)\underline{e} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\underline{e}^\perp$$

For rigid links, $\ddot{r} = \dot{r} = 0$.

① Rigid link AB.

B rel. A:

$$\begin{aligned}\ddot{\mathbf{r}}_B &= \ddot{\mathbf{r}}_A + (\ddot{x} - x\ddot{\theta})\mathbf{e}_r + (a\ddot{\theta} + \cancel{x\ddot{\theta}})\mathbf{e}_\theta \\ &= \ddot{\mathbf{r}}_A - a\ddot{\theta}\mathbf{e}_r + a\ddot{\theta}\mathbf{e}_\theta\end{aligned}$$

② Link AB w/ slider.

B rel. A:

$$\ddot{\mathbf{r}}_B = \ddot{\mathbf{r}}_A + (\ddot{x} - x\ddot{\theta})\mathbf{e}_r + (x\ddot{\theta} + 2\dot{x}\dot{\theta})\mathbf{e}_\theta$$

C rel. A:

$$\begin{aligned}\ddot{\mathbf{r}}_C &= \ddot{\mathbf{r}}_A + (\ddot{x} - a\ddot{\theta})\mathbf{e}_r + (a\ddot{\theta} + \cancel{2\dot{x}\dot{\theta}})\mathbf{e}_\theta \\ &= \ddot{\mathbf{r}}_A - a\ddot{\theta}\mathbf{e}_r + a\ddot{\theta}\mathbf{e}_\theta\end{aligned}$$

x rel. C:

$$\begin{aligned}\ddot{\mathbf{r}}_x &= \ddot{\mathbf{r}}_C + (\ddot{x} - x\ddot{\theta})\mathbf{e}_r + (x\ddot{\theta} + 2\dot{x}\dot{\theta})\mathbf{e}_\theta \\ &= \ddot{\mathbf{r}}_C + \ddot{x}\mathbf{e}_r + 2\dot{x}\dot{\theta}\mathbf{e}_\theta\end{aligned}$$

→ After drawing a pt. on the acceleration diagram, we can find other pts by considering their rel. accelerations rel. to the initial pt.

Inertia forces in mechanisms

- After calculating the accelerations of all components within the mechanism, we can add in distributed inertia forces and couples acting on the COM of links and sliders.
- The method of virtual power can be used to determine the driving torque needed to resist the inertia forces of the accelerating mechanism.
- * We can ignore components that are massless.

planar vectorial dynamics

Differentiation of vectors.

- the general result for differentiation of a vector $\mathbf{r}$ which is rotating at angular velocity ω is

$$\dot{\mathbf{r}} = \frac{d\mathbf{r}}{dt} + \omega \times \mathbf{r}$$

This works in 3D.

- In the special case of planar motion, we can define the position of B rel. to A is

$$\mathbf{r} = r\mathbf{e}$$

where $\mathbf{e}$ is a unit vector along AB.

If the angular velocity of AB is $\omega = \dot{\theta}\mathbf{k}$, then the velocity of B rel. to A is

$$\dot{\mathbf{r}} = \dot{r}\mathbf{e} + r\dot{\theta}\mathbf{e}^\perp$$

where $\mathbf{e}^\perp$ is a unit vector $\perp$ to $\mathbf{e}$.

the acceleration of B rel. to A is

$$\ddot{\mathbf{r}} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\mathbf{e}^\perp$$

special case — release from rest,

- In general, the acceleration of B rel. to A is

$$\ddot{\mathbf{r}} = (\ddot{\mathbf{r}} - \ddot{\mathbf{r}}_0)\mathbf{e} + (\ddot{\theta} + 2\dot{\theta}\dot{\mathbf{e}})\mathbf{e}^*$$

For a rigid link AB instantaneously at rest, then $\dot{\mathbf{r}}=0$, $\ddot{\mathbf{r}}=0$ and $\dot{\theta}=0$, so

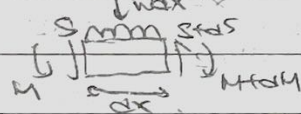
$$\ddot{\mathbf{r}} = \ddot{\theta}\mathbf{e}^*$$

- This dramatically simplifies the construction of an acceleration diagram

Bending moments and shear forces in mechanisms

Bending moments and shear forces

- Consider a beam subject to a distributed load $w(x)$.



Eqn. of the small element dx gives

$$w = \frac{dS}{dx}$$

$$S = \frac{dM}{dx}$$

Note that the max BM occurs where $S=0$.

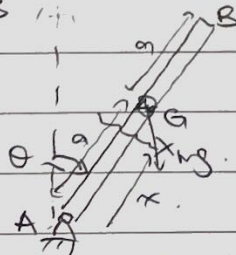
* Refer to structures notes on sketching SF/BM diagrams

Inertia forces in beams

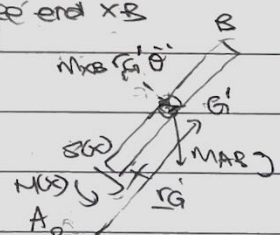
- Usually, when solving for the SF $S(x)$ or BM $M(x)$, we cut at a general pt. x and consider the FBD of the free end. ($S(x)$ found by NZL, $M(x)$ found by moments).

- e.g. A chimney of height $2a$ is being demolished. After an explosive charge goes off at its base, the chimney begins to rotate under the action of gravity. Find the location of max BM.

Whole chimney AB



Free end XB



cons. of mech. energy:

$$mg \Delta h = \frac{1}{2}mv^2 + \frac{1}{2}I_G\dot{\theta}^2$$

$$mg(1 - \cos\theta) = \frac{1}{2}m\dot{\theta}^2 + \frac{1}{2}(\frac{1}{3}m\dot{\theta}^2)$$

$$\frac{1}{2}\dot{\theta}^2 = \frac{3g}{2a}(1 - \cos\theta)$$

$$\dot{\theta} = \sqrt{\frac{3g}{a}(1 - \cos\theta)} = \sqrt{\frac{3g}{a}} \sin\frac{\theta}{2}$$

eqn. of free end XB: NZL $\uparrow$: $M_{XB} + G'_1\dot{\theta} = S(x) + M_{XB}g \sin\theta$

$$S(x) = M_{XB} \left[\left(\frac{x+2a}{2} \right) \frac{3g}{2a} \sin\theta - g \sin\theta \right]$$

For max BM, $S=0$, so

$$\left(\frac{x+2a}{2} \right) \frac{3g}{2a} \sin\theta = g \sin\theta$$

$$x = \frac{2a}{3}$$

3D kinematics and gyroscopic mechanics

Differentiation of rotating vectors in 3D.

- for planar motion, it is convenient to use $\mathbf{e}, \mathbf{e}^*$ notation. However in general, when $\mathbf{e}$ is not $\perp$ to Σ , the cross products for velocity and acceleration need to be handled in full.
- the velocity is obtained by differentiation:

$$\dot{\mathbf{r}} = \frac{d\mathbf{r}}{dt} + \boldsymbol{\omega} \times \mathbf{r}$$

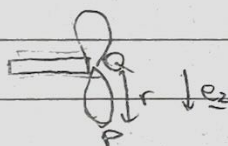
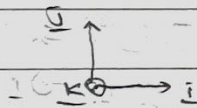
The acceleration is also obtained by differentiation:

$$\begin{aligned}\ddot{\mathbf{r}} &= \frac{d}{dt} \left[\frac{d\mathbf{r}}{dt} \right] + \frac{d}{dt} [\boldsymbol{\omega} \times \mathbf{r}] \\ &= \frac{d^2\mathbf{r}}{dt^2} + \boldsymbol{\omega} \times \frac{d\mathbf{r}}{dt} + \frac{d\boldsymbol{\omega}}{dt} \times \mathbf{r} + \boldsymbol{\omega} \times \frac{d\mathbf{r}}{dt} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})\end{aligned}$$

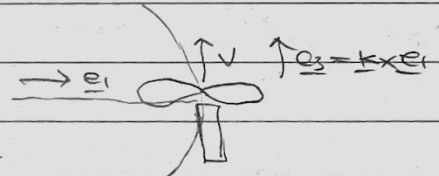
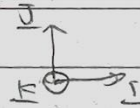
$$\ddot{\mathbf{r}} = \frac{d^2\mathbf{r}}{dt^2} + 2\boldsymbol{\omega} \times \frac{d\mathbf{r}}{dt} + \frac{d\boldsymbol{\omega}}{dt} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$$

- It is often simpler to find acceleration from first principles — the key is to keep track of the angular velocities and which unit vectors have which angular velocities.
- e.g. A single propeller aircraft is in steady level flight of speed v on a circular path of radius R . It has a propeller of radius r that is spinning at angular velocity ω as viewed by the pilot. At an instant, the tip P of the propeller blade is at its lowest pt. Find the instantaneous acceleration of P .

side view



top view

 $\mathbf{e}_1$ (and $\mathbf{e}_3$) rotates w (the plane) and $\mathbf{e}_2$ rotates the propeller.

$$\boldsymbol{\omega}_1 = \frac{v}{R} \mathbf{k}$$

$$\boldsymbol{\omega}_2 = \frac{v}{R} \mathbf{k} + \omega \mathbf{e}_2$$

$$\dot{\boldsymbol{\omega}}_1 = 0$$

$$\dot{\boldsymbol{\omega}}_2 = \dot{\omega} \mathbf{e}_2 = \omega \cdot \frac{v}{R} \mathbf{e}_1$$

The position, velocity and acceleration of P is.

$$\mathbf{r}_P = R\mathbf{e}_1 + r\mathbf{e}_2$$

$$\dot{\mathbf{r}}_P = R(\dot{\mathbf{e}}_1) + r(\dot{\mathbf{e}}_2)$$

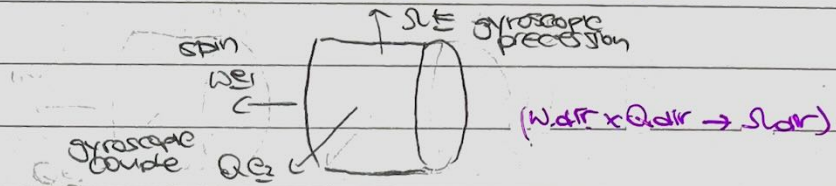
$$\ddot{\mathbf{r}}_P = R[(\dot{\boldsymbol{\omega}}_1 \times \mathbf{e}_1) + \boldsymbol{\omega}_1 \times (\boldsymbol{\omega}_1 \times \mathbf{e}_1)] + r[(\dot{\boldsymbol{\omega}}_2 \times \mathbf{e}_2) + \boldsymbol{\omega}_2 \times (\boldsymbol{\omega}_2 \times \mathbf{e}_2)]$$

At that instant, $\mathbf{e}_1 = \mathbf{i}$, $\mathbf{e}_2 = \mathbf{j}$ and $\mathbf{e}_3 = \mathbf{j}$

$$\begin{aligned}\therefore \ddot{\mathbf{r}}_P &= R \left[\omega + \left(\frac{v}{R} \mathbf{k} \right) \times \left(\frac{v}{R} \mathbf{k} \times \mathbf{i} \right) \right] + r \left[\left(-\frac{v\omega}{R} \mathbf{i} \times \mathbf{j} \right) + \left(\frac{v}{R} \mathbf{k} + \omega \mathbf{j} \right) \times \left(\left(\frac{v}{R} \mathbf{k} + \omega \mathbf{j} \right) \times \mathbf{j} \right) \right] \\ &= -\frac{v^2}{R} \mathbf{i} - \frac{r\omega v}{R} \mathbf{j} + r \left[\left(\frac{v}{R} \mathbf{k} + \omega \mathbf{j} \right) \times (-\omega \mathbf{i}) \right] \\ &= -\frac{v^2}{R} \mathbf{i} - \frac{2r\omega v}{R} \mathbf{j} + r\omega^2 \mathbf{k}\end{aligned}$$

Gyroscopic mechanics

- Consider a rigid body rotating w/ angular velocity $\omega \mathbf{e}_1$. If we apply a torque $\mathbf{Q} \perp$ to the angular velocity $\omega \mathbf{e}_1$, it precesses w/ an angular velocity $\Omega \mathbf{e}_2$.



The angular momentum $\mathbf{h}$ is given by

$$\mathbf{h} = J \omega \mathbf{e}_1$$

Applying N2L of rotation, we have $\frac{d\mathbf{h}}{dt} = 0$ here since applied torque $\perp$ to angular momentum

$$\mathbf{Q} = \dot{\mathbf{h}} = \frac{d\mathbf{h}}{dt} + \Omega \mathbf{e}_2 \times \mathbf{h} = \Omega \mathbf{e}_2 \times (J \omega \mathbf{e}_1)$$

$$\mathbf{Q} = J \Omega \omega \mathbf{e}_2$$

$\mathbf{Q}$ is the gyroscopic couple and Ω is the gyroscopic precession.

Lagrangian dynamics

Generalised coordinates.

- The no. of quantities we need to describe a mechanism's configuration is the mechanism's no. of degree of freedom.
- To describe an n degree of freedom mechanism, we can use its generalised coords, a set of n variables, denoted as $\{q_i\} = q_1, q_2, \dots, q_n$ or $q = (q_1, q_2, \dots, q_n)$
- * q can represent positions, or angles.

Lagrangian mechanics.

- consider a mechanical system w/ n generalised coords, $q_1, q_2, \dots, q_n$.

We calculate the kinetic energy T and potential energy V in terms of q_i and $\dot{q}_i$ Then, we calculate the system's Lagrangian L , as a function of $q_i, \dot{q}_i$ and time t , as

$$L(q_1, q_2, \dots, \dot{q}_1, \dot{q}_2, \dots, t) = T - V \quad \leftarrow \text{not the total energy } T+V$$

- For each generalised coord q_i , we can calculate the generalised momentum p_i and generalised force F_i by taking the derivatives of the Lagrangian.

$$p_i = \frac{\partial L}{\partial \dot{q}_i}$$

$$F_i = \frac{\partial L}{\partial q_i}$$

(must be partial derivatives! e.g. $\frac{\partial}{\partial q_1}(q_1^2 q_2 \dot{q}_1) = 2q_1 q_2 \dot{q}_1$)

The eqn. of motion for each generalised coordinate is then simply

$$\frac{d}{dt}(p_i) = F_i$$

(must be full derivative! e.g. $\frac{d}{dt}(q_1 \dot{q}_1) = \dot{q}_1^2 + q_1 \ddot{q}_1$)

- In terms of the Lagrangian only, the eqn. of motion for q_i can be written

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i}$$

$$\text{or} \quad \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0$$

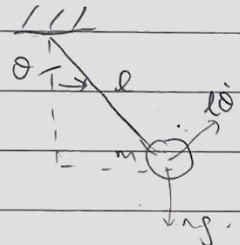
Example - simple pendulum.

- consider a simple pendulum

$$T = \frac{1}{2}m(\ell\dot{\theta})^2 \quad V = -mg\ell\cos\theta \quad \rightarrow \quad L = T - V = \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell\cos\theta$$

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = m\ell\dot{\theta} \quad F_\theta = \frac{\partial L}{\partial \theta} = -mg\ell\sin\theta$$

$$\therefore \frac{d}{dt}(m\ell\dot{\theta}) = -mg\ell\sin\theta \quad \rightarrow \quad \ddot{\theta} = -\frac{g}{\ell}\sin\theta$$



Example - bead on a spinning hoop.

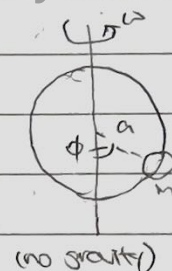
- consider a bead on a spinning hoop, as illustrated.

$$V = (a\dot{\phi})^2 + (a\sin\phi\omega)^2 = a^2\dot{\phi}^2 + a^2\sin^2\phi\omega^2$$

$$T = \frac{1}{2}m(a^2\dot{\phi}^2 + a^2\sin^2\phi\omega^2) \quad V=0 \rightarrow L = T - V = \frac{1}{2}m(a^2\dot{\phi}^2 + a^2\sin^2\phi\omega^2)$$

$$P_\phi = \frac{\partial L}{\partial \dot{\phi}} = m a^2 \dot{\phi} \quad F_\phi = m a^2 \omega^2 \sin\phi \cos\phi$$

$$\therefore \frac{d}{dt}(m a^2 \dot{\phi}) = m a^2 \omega^2 \sin\phi \cos\phi \rightarrow \ddot{\phi} = \omega^2 \sin\phi \cos\phi$$



Example - two simple pendulums.

- consider two entirely independent pendulums, w/ diff. length, mass, angle.

The Lagrangian $L = T - V$ is the sum of the individual Lagrangians for each pendulum.

$$L(\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2) = T_1 + T_2 - V_1 - V_2 = (T_1 - V_1) + (T_2 - V_2) = L_1(\theta_1, \dot{\theta}_1) + L_2(\theta_2, \dot{\theta}_2)$$

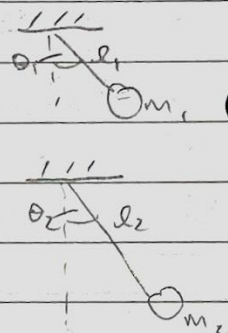
$$\text{where } L_1(\theta_1, \dot{\theta}_1) = \frac{1}{2}m_1 l_1 \dot{\theta}_1^2 + m_1 g l_1 \cos\theta_1; \quad L_2(\theta_2, \dot{\theta}_2) = \frac{1}{2}m_2 l_2 \dot{\theta}_2^2 + m_2 g l_2 \cos\theta_2$$

If we work out the generalised momentum and force for θ_i , we get the answer we would expect from just using L_i

$$P_{\theta_i} = \frac{\partial L}{\partial \dot{\theta}_i} = \frac{\partial L_i}{\partial \dot{\theta}_i} = m_i l_i^2 \ddot{\theta}_i \quad F_{\theta_i} = \frac{\partial L}{\partial \theta_i} = \frac{\partial L_i}{\partial \theta_i} = -m_i g l_i \sin\theta_i$$

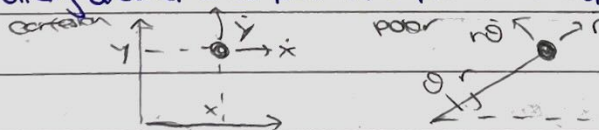
$$\therefore \frac{d}{dt}(m_i l_i^2 \ddot{\theta}_i) = -m_i g l_i \sin\theta_i \rightarrow \ddot{\theta}_i = -\frac{g}{l_i} \sin\theta_i$$

→ The Lagrangian for independent / uncoupled systems simply add together, and the motion for each subsystem will then be governed by its sub-Lagrangian.



Example - particle in 2D

- consider a particle sliding around a 2D plane in a potential energy V



$$\text{cartesian: } T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) \quad V = V(x, y) \rightarrow L = T - V = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - V(x, y)$$

$$P_x = \frac{\partial L}{\partial \dot{x}} = m\dot{x} \quad F_x = \frac{\partial L}{\partial x} = -\frac{\partial V}{\partial x} \rightarrow \frac{d}{dt}(m\dot{x}) = -\frac{\partial V}{\partial x}$$

$$P_y = \frac{\partial L}{\partial \dot{y}} = m\dot{y} \quad F_y = \frac{\partial L}{\partial y} = -\frac{\partial V}{\partial y} \rightarrow \frac{d}{dt}(m\dot{y}) = -\frac{\partial V}{\partial y}$$

$$\therefore m\ddot{x} = -\frac{\partial V}{\partial x}, \quad m\ddot{y} = -\frac{\partial V}{\partial y}$$

$$\text{polar: } T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) \quad V = V(r, \theta) \rightarrow L = T - V = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r, \theta)$$

$$P_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r} \quad F_r = \frac{\partial L}{\partial r} = m r \dot{\theta}^2 - \frac{\partial V}{\partial r} \rightarrow \frac{d}{dt}(m\dot{r}) = m r \dot{\theta}^2 - \frac{\partial V}{\partial r}$$

$$P_\theta = \frac{\partial L}{\partial \dot{\theta}} = m r^2 \dot{\theta} \quad F_\theta = \frac{\partial L}{\partial \theta} = -\frac{\partial V}{\partial \theta} \rightarrow \frac{d}{dt}(m r^2 \dot{\theta}) = -\frac{\partial V}{\partial \theta}$$

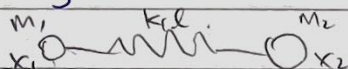
$$\therefore m\ddot{r} = m r \dot{\theta}^2 - \frac{\partial V}{\partial r}, \quad m(r^2 \ddot{\theta} + 2r\dot{r}\dot{\theta}) = -\frac{\partial V}{\partial \theta}$$

$$m(\ddot{r} - r\dot{\theta}^2) = -\frac{\partial V}{\partial r}, \quad m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = -\frac{1}{r}\frac{\partial V}{\partial \theta}$$

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Example - two masses connected by a spring.

- Consider two masses, m_1 and m_2 , each moving on the x -axis, w/ coords x_1 and x_2 , connected by a spring of natural length and spring const. k .



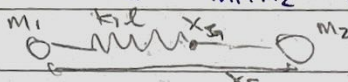
$$T = \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 \quad V = \frac{1}{2}k(x_2 - x_1 - l)^2 \rightarrow L = T - V = \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 - \frac{1}{2}k(x_2 - x_1 - l)^2$$

$$p_{x_1} = \frac{\partial L}{\partial \dot{x}_1} = m_1\dot{x}_1 \quad F_{x_1} = \frac{\partial L}{\partial x_1} = k(x_2 - x_1 - l) \rightarrow \frac{d}{dt}(m_1\dot{x}_1) = k(x_2 - x_1 - l)$$

$$p_{x_2} = \frac{\partial L}{\partial \dot{x}_2} = m_2\dot{x}_2 \quad F_{x_2} = \frac{\partial L}{\partial x_2} = -k(x_2 - x_1 - l) \rightarrow \frac{d}{dt}(m_2\dot{x}_2) = -k(x_2 - x_1 - l)$$

$$\therefore m_1\ddot{x}_1 = k(x_2 - x_1 - l) \quad m_2\ddot{x}_2 = -k(x_2 - x_1 - l)$$

Now consider using a diff. set of coords, $x_G = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$, $x_S = x_2 - x_1$



$$T = \frac{1}{2}(m_1 + m_2)\dot{x}_G^2 + \frac{1}{2}\frac{m_1m_2}{m_1 + m_2}\dot{x}_S^2 \quad V = \frac{1}{2}k(x_S - l)^2 \rightarrow L = T - V = \frac{1}{2}(m_1 + m_2)\dot{x}_G^2 + \frac{1}{2}\frac{m_1m_2}{m_1 + m_2}\dot{x}_S^2 - \frac{1}{2}k(x_S - l)^2$$

* Notice the Lagrangian has turned into the sum of two separate parts, $x_G, x_S \rightarrow$ uncoupled eqn. of motion

$$p_{x_G} = \frac{\partial L}{\partial \dot{x}_G} = (m_1 + m_2)\dot{x}_G \quad F_{x_G} = \frac{\partial L}{\partial x_G} = 0 \rightarrow \frac{d}{dt}((m_1 + m_2)\dot{x}_G) = 0$$

$$p_{x_S} = \frac{\partial L}{\partial \dot{x}_S} = \frac{m_1m_2}{m_1 + m_2}\dot{x}_S \quad F_{x_S} = \frac{\partial L}{\partial x_S} = -k(x_S - l) \rightarrow \frac{d}{dt}\left(\frac{m_1m_2}{m_1 + m_2}\dot{x}_S\right) = -k(x_S - l)$$

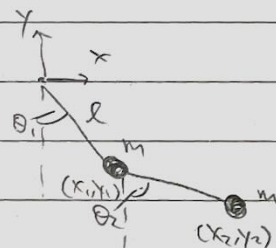
$$\therefore (m_1 + m_2)\ddot{x}_G = 0 \quad \frac{m_1m_2}{m_1 + m_2}\ddot{x}_S = -k(x_S - l)$$

Example - double pendulum.

- Consider the double pendulum. The cartesian positions of the two masses are

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} l \sin \theta_1 \\ -l \cos \theta_1 \end{bmatrix} \quad \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} l \sin \theta_1 + l \sin \theta_2 \\ -l \cos \theta_1 - l \cos \theta_2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} \dot{x}_1 \\ \dot{y}_1 \end{bmatrix} = \begin{bmatrix} l \cos \theta_1 \cdot \dot{\theta}_1 \\ l \sin \theta_1 \cdot \dot{\theta}_1 \end{bmatrix} \quad \begin{bmatrix} \dot{x}_2 \\ \dot{y}_2 \end{bmatrix} = \begin{bmatrix} l \cos \theta_1 \cdot \dot{\theta}_1 + l \cos \theta_2 \cdot \dot{\theta}_2 \\ l \sin \theta_1 \cdot \dot{\theta}_1 + l \sin \theta_2 \cdot \dot{\theta}_2 \end{bmatrix}$$



$$T = \frac{1}{2}m(\dot{x}_1^2 + \dot{y}_1^2 + \dot{x}_2^2 + \dot{y}_2^2) = \frac{1}{2}m(l^2\dot{\theta}_1^2 + l^2\dot{\theta}_2^2 + 2l^2\dot{\theta}_1\dot{\theta}_2(\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2))$$

$$V = mgy_1 + mgy_2 = -mgl(2\cos\theta_1 + \cos\theta_2)$$

$$\therefore L = T - V = \frac{1}{2}ml^2(2\dot{\theta}_1^2 + \dot{\theta}_2^2 + 2\dot{\theta}_1\dot{\theta}_2\cos(\theta_2 - \theta_1)) + mgl(2\cos\theta_1 + \cos\theta_2)$$

$$p_{\theta_1} = \frac{\partial L}{\partial \dot{\theta}_1} = ml^2(2\dot{\theta}_1 + \dot{\theta}_2\cos(\theta_2 - \theta_1)) \quad F_{\theta_1} = \frac{\partial L}{\partial \theta_1} = ml^2\dot{\theta}_1\dot{\theta}_2\sin(\theta_2 - \theta_1) - 2mgl\sin\theta_1$$

$$p_{\theta_2} = \frac{\partial L}{\partial \dot{\theta}_2} = ml^2(\dot{\theta}_2 + \dot{\theta}_1\cos(\theta_2 - \theta_1)) \quad F_{\theta_2} = \frac{\partial L}{\partial \theta_2} = -ml^2\dot{\theta}_1\dot{\theta}_2\sin(\theta_2 - \theta_1) - mgl\sin\theta_2$$

$$\rightarrow \frac{d}{dt}(p_{\theta_1}) = ml^2(2\ddot{\theta}_1 + \ddot{\theta}_2\cos(\theta_2 - \theta_1) - \dot{\theta}_2\sin(\theta_2 - \theta_1)(\dot{\theta}_2 - \dot{\theta}_1)) = ml^2\ddot{\theta}_1\dot{\theta}_2\sin(\theta_2 - \theta_1) - 2mgl\sin\theta_1$$

$$\frac{d}{dt}(p_{\theta_2}) = ml^2(\ddot{\theta}_2 + \ddot{\theta}_1\cos(\theta_2 - \theta_1) - \dot{\theta}_1\sin(\theta_2 - \theta_1)(\dot{\theta}_2 - \dot{\theta}_1)) = -ml^2\dot{\theta}_1\dot{\theta}_2\sin(\theta_2 - \theta_1) - mgl\sin\theta_2$$

Ticking the eqns up, we get.

$$\begin{cases} 2\ddot{\theta}_1 + \ddot{\theta}_2\cos(\theta_2 - \theta_1) - \dot{\theta}_2^2\sin(\theta_2 - \theta_1) = -2\frac{g}{l}\sin\theta_1 \\ \ddot{\theta}_2 + \ddot{\theta}_1\cos(\theta_2 - \theta_1) + \dot{\theta}_1^2\sin(\theta_2 - \theta_1) = -\frac{g}{l}\sin\theta_2 \end{cases}$$

Conservation laws

Noether's theorem.

- In general, the Lagrangian eqn. of motion is

$$\frac{d}{dt}(p_i) = F_i$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i}$$

If the Lagrangian L does not depend on q_i , then the generalised force $F_i = \frac{\partial L}{\partial q_i} = 0$, and the eqn becomes

$$\frac{d}{dt}(p_i) = 0$$

which simply states that the generalised momentum p_i does not change during the motion, i.e. it is const./conserved.

- In summary, Noether's thm. states, if the Lagrangian doesn't depend on a generalised coord q_i , then the generalised momentum p_i is conserved. (symmetry/invariance $\leftrightarrow$ conservation)

Conservation laws.

- Lagrangian invariant in translation $\rightarrow$ consv. of linear momentum

Lagrangian invariant in rotation $\rightarrow$ consv. of angular momentum

Lagrangian invariant in time $\rightarrow$ consv. of energy.

One degree of freedom mechanics

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One degree of freedom systems

One degree of freedom systems.

- The Lagrangian of a mechanism w/ only one degree of freedom q will be of the form.

$$L(q, \dot{q}) = \frac{1}{2} m \dot{q}^2 - V(q)$$

- The Lagrangian gives rise to a single generalised momentum and force.

$$p = \frac{\partial L}{\partial \dot{q}} = m \dot{q}$$

$$F = \frac{\partial L}{\partial q} = -V'(q).$$

so the eqn. of motion is simply

$$\frac{d}{dt}(m \dot{q}) = m \ddot{q} = -V'(q)$$

- To solve for the motion of the system, we need to integrate twice: $\ddot{q} \rightarrow \dot{q} \rightarrow q$.

First integral of the equation of motion.

- The first integration can always be done analytically. The Lagrangian doesn't have any t dependence, so the motion will conserve total energy.

$$E = T + V = \frac{1}{2} m \dot{q}^2 + V(q).$$

To verify this for the one dof case, we multiply the eqn. of motion by $\dot{q}$

$$m \ddot{q} \dot{q} + V'(q) \dot{q} = 0$$

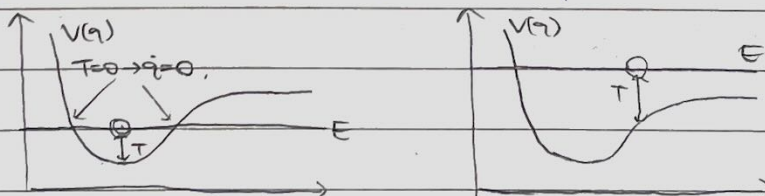
$$\frac{d}{dt} \left(\frac{1}{2} m \dot{q}^2 + V(q) \right) = 0$$

$$\therefore \frac{1}{2} m \dot{q}^2 + V(q) = E = \text{const.}$$

We can rearrange this to find an expression for $\dot{q}$.

$$\dot{q} = \sqrt{\frac{2}{m}(E - V(q))}$$

- If we draw a graph of $V(q)$, we can represent the total energy of a motion E as a straight horizontal line. At any position, the gap between the line and the potential gives the kinetic energy of the particle T , from which we can get the velocity



- Either the E line intersects the V curve twice $\rightarrow$ these are turning pts where $\dot{q}=0$
 $\rightarrow$ mechanism moves between the two turning pts periodically forever.
- Or E might be above the V curve (or intersect only once) $\rightarrow q$ will never turn around and eventually reach infinity.

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Equilibrium states

- In general, to find eqm. pts, simply set $q=q_0$ and $\dot{q}=\ddot{q}=0$ in the eqn of motion and see if there are any sol's.

- setting $\dot{q}=0$ in the original eqn. of motion, we see the eqm. pt. must satisfy

$$V'(q_0) = 0$$

i.e. it must be a stationary pt. of the potential energy $V(q)$.

Setting $\dot{q}=0$ in the energy expression, we see the eqm. pt. must also satisfy

$$E = V(q_0)$$

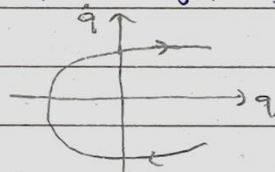
i.e. it must have zero kinetic energy T and thus be stationary.

→ eqmpts occur when a system is at maxima/minima of potential energy and is not moving.

Phase portrait

- A phase portrait is a plot of $\dot{q}$ against q , so each pt. corresponds to the mechanism being in a certain configuration and moving at a certain rate.

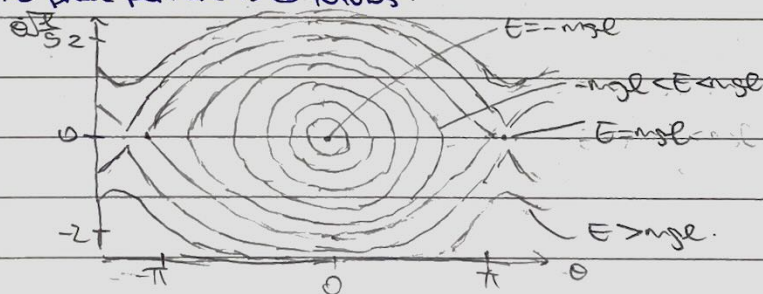
- As the mechanism moves, both q and $\dot{q}$ will change, tracing out a curve for the motion (orbit).



▲ The plot is self-consistent: q increases/decreases when $\dot{q}$ is pos/neg. → no need arrows.

- In general, there is one such line on the phase portrait for each value of the total energy E , which plots the motion of the mechanism when it moves w/ that particular value of E .

- For a simple pendulum, the phase portrait is as follows:



stability and instability

- When the system is a small distance from an eqn pt, $q = q_0 + \delta q$, the eqn. of motion $m\ddot{q} = -V'(q)$ becomes,

$$m\delta\ddot{q} = -V'(q_0 + \delta q)$$

Taking the first non-zero term from the Taylor expansion of $V'(q_0 + \delta q)$, we get.

$$m\delta\ddot{q} = -V'(q_0) - V''(q_0)\delta q + \dots$$

$$\therefore m\delta\ddot{q} = -V''(q_0)\delta q$$

which is an eqn. of motion for δq w/ a diff. character depending on the sign of $V''(q_0)$.

- If $V(q_0)$ is a min, $V''(q_0)$ is +ve $\rightarrow$ force pulls back to eqn. and is the eqn. of SHM (stable eqn).

$$\delta\ddot{q} = -\frac{V''(q_0)}{m}\delta q \rightarrow \delta q = a_0 \cos\left(\sqrt{\frac{V''(q_0)}{m}}t + \phi\right)$$

if δq starts small, it remains small $\rightarrow$ approx. eqn. of motion remains valid.

- If $V(q_0)$ is a max, $V''(q_0)$ is -ve $\rightarrow$ force pushes away from eqn and the eqn. has exponentially growing and decaying solns (unstable eqn).

$$\delta\ddot{q} = \frac{V''(q_0)}{m}\delta q \rightarrow \delta q = Ae^{\sqrt{\frac{V''(q_0)}{m}}t} + Be^{-\sqrt{\frac{V''(q_0)}{m}}t}$$

if δq starts small, it grows until it gets big and the Taylor series approx breaks down.

small vibrations analysis.

- For a system that is a small distance from a stable eqn. pt, $q = q_0 + \delta q$, the eqn. of motion for δq is

$$m\delta\ddot{q} = -V''(q_0)\delta q$$

The small perturbation in the generalised coord δq varies sinusoidally w/ angular frequency ω ,

$$\omega = \sqrt{\frac{V''(q_0)}{m}}$$

second integral of equation of motion

- We start w/ the expression for $\dot{q}$ from the first integral.

$$\dot{q} = \sqrt{\frac{2}{m}(E - V(q))}$$

Dividing by the RHS and integrating wrt q ,

$$\int \frac{\sqrt{m/2}}{E - V(q)} dq = \int dt = t$$

so if we start from q_i , and want to know when the particle reaches q_f , we simply evaluate.

$$t_{q_i \rightarrow q_f} = \int_{q_i}^{q_f} \frac{\sqrt{m/2}}{E - V(q)} dq$$

$\rightarrow$ all energy-conserving one dof motion can be reduced to a single final integral.

(However, the integral is rarely tractable analytically.)

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Reductible multiple degree of freedom problems.

Effective potential

- We can use consv. laws to effectively reduce the no. of dof for a system $\rightarrow$ turn the problem into a one dof problem. $\boxed{m\ddot{q} = -V_{\text{eff}}'(q)}$ [e.g. $m\ddot{r} = -V_{\text{eff}}'(r)$; $m r^2 \ddot{\theta} = -V_{\text{eff}}'(\theta)$]

- The overall strategy is:

$\hookrightarrow$ 1) Find the Lagrangian, and we should find L does not depend on q_k . $\rightarrow F_k = \frac{\partial L}{\partial q_k} = 0$
 $\rightarrow \frac{d}{dt}(p_k) = 0 \rightarrow p_k = h = \text{const.}$

$\hookrightarrow$ 2) Rearrange to express $\dot{q}_k, \ddot{q}_k$ in terms of h , then substitute into the eqn. of motion for the other generalised coord q_i .

$\hookrightarrow$ 3) We get an eqn. in the form $m\ddot{q}_i = -V_{\text{eff}}'(q_i)$, so integrate RHS wrt q_i to find $V_{\text{eff}}(q_i)$

- For applications where $q_k = \theta$, $q_i = r$, we can analyse the system further:

$\hookrightarrow$ 1) Set $V_{\text{eff}}'(q_i) = 0$ to find $q_i = q_{\text{min}}$ which minimises $V_{\text{eff}}(q_i)$, then substitute into the expression for $\dot{q}_k$ to find the angular speed at the eqm. pt.

$\hookrightarrow$ 2) Apply small vibration analysis to analyse motion when $q_i = q_{\text{min}} + \delta q_i$, then calculate the angular frequency of these perturbations.

Satellite motion.

- For a satellite going around Earth, the Lagrangian is given by

$$L(r, \theta, \dot{r}, \dot{\theta}) = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) + \frac{GMm}{r} \quad \begin{matrix} v = v(r) \\ \text{central potential} \end{matrix}$$

The generalised momenta and forces are

$$p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r}$$

$$F_r = \frac{\partial L}{\partial r} = m r \dot{\theta}^2 - \frac{GMm}{r^2}$$

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = m r^2 \dot{\theta}$$

$$F_\theta = \frac{\partial L}{\partial \theta} = 0 \quad \text{key!}$$

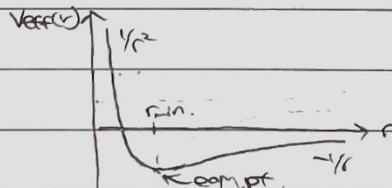
θ : $\frac{d}{dt}(m r^2 \dot{\theta}) = 0 \rightarrow m r^2 \dot{\theta} = h = \text{const.}$, i.e. angular momentum is conserved.

$$r$$
: $\frac{d}{dt}(m\dot{r}) = m r \dot{\theta}^2 - \frac{GMm}{r^2} \rightarrow m\ddot{r} = m r \left(\frac{h}{m r^2}\right)^2 - \frac{GMm}{r^2} = \frac{h^2}{m r^3} - \frac{GMm}{r^2}$

$$\therefore -V_{\text{eff}}'(r) = \frac{h^2}{m r^3} - \frac{GMm}{r^2} \rightarrow V_{\text{eff}}(r) = \int -\left(\frac{h^2}{m r^3} - \frac{GMm}{r^2}\right) dr = \frac{h^2}{2 m r^2} - \frac{GMm}{r}$$

centrifugal potential KE associated in the θ dir.

- sketching the effective potential as a function of r .



$V_{\text{eff}}(r)$ has a minima $\rightarrow$ stable eqm. pt at $r = r_{\text{min}}$.

$$V_{\text{eff}}'(r) = -\frac{h^2}{m r^3} + \frac{GMm}{r^2} = 0 \rightarrow r_{\text{min}} = \frac{h^2}{GMm}$$

The angular speed at the eqm. pt. $\dot{\theta}$ is

$$\dot{\theta} = \frac{h}{m r_{\text{min}}^2} = \sqrt{\frac{GM}{r_{\text{min}}}}$$

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- Applying small vibration analysis, when $r = r_{min} + \delta r$, the eqn. of motion for δr is

$$m \delta \ddot{r} = -V_{eff}''(r_{min}) \delta r$$

$$V_{eff}''(r) = \frac{3h^2}{mr^4} - \frac{2GMm}{r^3} \rightarrow V_{eff}''(r_{min}) = \frac{3(GMmr_{min})}{r_{min}^4} - \frac{2GMm}{r_{min}^3} = \frac{GMm}{r_{min}^3}$$

$$\therefore \delta \ddot{r} = -\frac{GM}{r_{min}^3} \delta r$$

so the perturbation in δr varies sinusoidally w/ angular frequency ω .

$$\omega = \sqrt{\frac{GM}{r_{min}^3}}$$

- We find that $\dot{\theta} = \omega \rightarrow$ satellite does one complete radial oscillation on each circular orbit $\rightarrow$ orbit closes.

* If the satellite has more energy, small vibration analysis no longer applies, but the orbit remains closed and elliptical.

Motion in a cone

- Consider a particle sliding along the surface of a cone under gravity

$$T = \frac{1}{2} m (\dot{r}^2 + \dot{\phi}^2) = \frac{1}{2} m (\dot{r}^2 + (\frac{\dot{\phi}}{\sin \theta})^2) \quad V = -mgz = -mg r \cos \theta \quad V = V(r, \phi) \text{ central potential}$$

$$\therefore L(r, \phi, \dot{r}, \dot{\phi}) = \frac{1}{2} m (\dot{r}^2 + \dot{\phi}^2 \cot^2 \theta) - mgr \cos \theta$$

The generalised momenta and forces are

$$P_r = \frac{\partial L}{\partial \dot{r}} = m \dot{r} \cos^2 \theta$$

$$F_r = \frac{\partial L}{\partial r} = m \dot{\phi}^2 \cot^2 \theta - mg \cos \theta$$

$$P_\phi = \frac{\partial L}{\partial \dot{\phi}} = m r^2 \dot{\phi} \cot^2 \theta$$

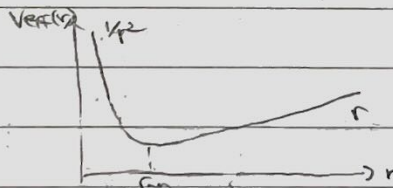
$$F_\phi = \frac{\partial L}{\partial \phi} = 0 \text{ kept.}$$

$$\phi: \frac{d}{dt} (m r^2 \dot{\phi} \cot^2 \theta) = 0 \rightarrow m r^2 \dot{\phi} = h = \text{const.} \quad \text{i.e. angular momentum is conserved.}$$

$$r: \frac{d}{dt} (m \dot{r} \cos^2 \theta) = m \ddot{r} \cos^2 \theta - m g \sin \theta \cos \theta \rightarrow m \ddot{r} = m r \left(\frac{h}{m r^2} \right)^2 \sin^2 \theta - m g \sin \theta \cos \theta = \frac{h^2 \sin^2 \theta}{m r^3} - m g \sin \theta \cos \theta$$

$$\therefore -V_{eff}'(r) = \frac{h^2 \sin^2 \theta}{m r^3} - m g \sin \theta \cos \theta \rightarrow V_{eff}(r) = \int -\left(\frac{h^2 \sin^2 \theta}{m r^3} - m g \sin \theta \cos \theta \right) dr = \frac{h^2 \sin^2 \theta}{2 m r^2} + m g r \sin \theta \cos \theta$$

- sketching the effective potential as a function of r



$V_{eff}(r)$ has a minima $\rightarrow$ stable eqn. pt. at $r = r_{min}$.

$$V_{eff}'(r) = -\frac{h^2 \sin^2 \theta}{m r^3} + m g \sin \theta \cos \theta = 0 \rightarrow r_{min}^3 = \frac{h^2 \sin^2 \theta}{m g \cos \theta}$$

The angular speed at the eqn. pt. ϕ is

$$\dot{\phi} = \frac{h}{m r_{min}^2} = \sqrt{\frac{g}{r_{min} \cos \theta}}$$

- Applying small vibration analysis, when $r = r_{min} + \delta r$, the eqn. of motion for δr is

$$m \delta \ddot{r} = -V_{eff}''(r_{min}) \delta r$$

$$V_{eff}''(r) = \frac{3h^2 \sin^2 \theta}{m r^4} \rightarrow V_{eff}''(r_{min}) = \frac{3 m g \sin \theta \cos \theta}{r_{min}}$$

$$\therefore \delta \ddot{r} = -\frac{3 g \sin \theta \cos \theta}{r_{min}} \delta r$$

so the perturbation in δr varies sinusoidally w/ angular frequency ω .

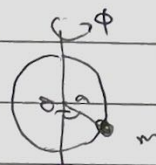
$$\omega = \sqrt{\frac{3 g \sin \theta \cos \theta}{r_{min}}}$$

- We find that $\dot{\theta} = 0$ only when $\sin \theta = 1/\sqrt{3}$, i.e. $\theta = 35^\circ$, or else it is just quasi-periodic.

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Bead on a freely rotating hoop

- consider a bead that can slide around a hoop, which is free to spin around a diameter. The Lagrangian is given by



$$L(\theta, \phi, \dot{\theta}, \dot{\phi}) = \frac{1}{2}m(a^2\dot{\theta}^2 + a^2\sin^2\theta\dot{\phi}^2)$$

The generalised momenta and forces are

$$p_{\theta} = \frac{\partial L}{\partial \dot{\theta}} = ma^2\dot{\theta}$$

$$p_{\phi} = \frac{\partial L}{\partial \dot{\phi}} = ma^2\sin^2\theta\dot{\phi}$$

$$F_{\theta} = \frac{\partial L}{\partial \theta} = ma^2\sin\theta\dot{\phi}^2$$

$$F_{\phi} = \frac{\partial L}{\partial \phi} = 0 \quad \text{key!}$$

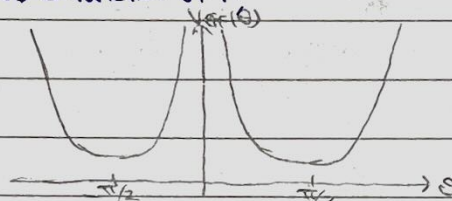
$$\phi: \frac{d}{dt}(ma^2\sin^2\theta\dot{\phi}) = 0 \rightarrow ma^2\sin^2\theta\dot{\phi} = h = \text{const}, \text{ i.e. angular momentum is conserved.}$$

$$\theta: \frac{d}{dt}(ma^2\dot{\theta}) = ma^2\sin\theta\cos\theta\dot{\phi}^2 \rightarrow m\ddot{\theta} = ma^2\sin\theta\cos\theta\left(\frac{h}{ma^2\sin^2\theta}\right)^2 = \frac{h^2\cos\theta}{ma^2\sin^3\theta}$$

We write an eqn. in the form $ma^2\ddot{\theta} = -V_{\text{eff}}'(\theta)$, so

$$-V_{\text{eff}}'(\theta) = \frac{h^2\cos\theta}{ma^2\sin^3\theta} \rightarrow V_{\text{eff}}(\theta) = \int -\left(\frac{h^2\cos\theta}{ma^2\sin^3\theta}\right) d\theta = \frac{h^2}{2ma^2\sin^2\theta}$$

- Sketching the effective potential as a function of θ



the minima of this potential is at $\theta = \pm\pi/2$.

Two body problem

- A two body problem can be reduced to a one dot problem by using special coords $\vec{r}_1, \vec{r}_2$.
- consider two particles, m_1 and m_2 w/ positions $\vec{r}_1 = (x_1, y_1, z_1)$ and $\vec{r}_2 = (x_2, y_2, z_2)$. If the two particles exert a force on each other, the potential will be a function of separation $|\vec{r}_1 - \vec{r}_2|$.

$$\therefore L(\vec{r}_1, \vec{r}_2, \dot{\vec{r}}_1, \dot{\vec{r}}_2) = \frac{1}{2}m_1|\dot{\vec{r}}_1|^2 + \frac{1}{2}m_2|\dot{\vec{r}}_2|^2 - V(|\vec{r}_1 - \vec{r}_2|)$$

choosing the generalised coords to $\vec{r}_1 = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}$ and $\vec{r} = \vec{r}_2 - \vec{r}_1$, we have.

$$\vec{r}_1 = \vec{r}_1 - \frac{m_2}{m_1 + m_2}\vec{r}, \quad \vec{r}_2 = \vec{r}_1 + \frac{m_1}{m_1 + m_2}\vec{r}$$

In terms of the new coords, the Lagrangian is now

$$L(\vec{r}_1, \vec{r}, \dot{\vec{r}}_1, \dot{\vec{r}}) = \frac{1}{2}(m_1 + m_2)|\dot{\vec{r}}_1|^2 + \frac{1}{2}M|\dot{\vec{r}}|^2 - V(|\vec{r}|)$$

where $M = \frac{m_1 m_2}{m_1 + m_2}$ is the reduced mass.

the Lagrangian is the sum of two independent sub-Lagrangians (uncoupled motion)

$$L_1 = \frac{1}{2}(m_1 + m_2)|\dot{\vec{r}}_1|^2$$

$$L_2 = \frac{1}{2}M|\dot{\vec{r}}|^2 - V(|\vec{r}|)$$

and the eqn. of motion for $\vec{r}_1$ turns out to be cons. of total momentum

$$\frac{d}{dt}((m_1 + m_2)\dot{\vec{r}}_1) = 0$$

Stability and normal modes

General normal mode treatment.

- consider a general n dof. mechanism, w/ generalised coords $q_1, q_2, \dots, q_n$. Let $\mathbf{q} = (q_1, q_2, \dots, q_n)$.
- the kinetic energy is quadratic in $\dot{q}_j$, so it always has to form

$$T = \sum_{i,j} \frac{1}{2} M_{ij}(\mathbf{q}) \dot{q}_i \dot{q}_j = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}}$$

where $\mathbf{M}(\mathbf{q})$ is a matrix of mass-like quantities w/ elements M_{ij} .(e.g. in polar coords, a mass m has $T = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \dot{\theta}^2 = \frac{1}{2} [\dot{r} \ \dot{\theta}] \begin{bmatrix} m & 0 \\ 0 & m r^2 \end{bmatrix} \begin{bmatrix} \dot{r} \\ \dot{\theta} \end{bmatrix}$).& we can always choose $\mathbf{M}(\mathbf{q})$ to be symmetric, as an antisymmetric part makes no contribution to T .

- In general, the potential energy will not be quadratic, but it won't depend on velocity, i.e.

$$V = V(\mathbf{q})$$

- The Lagrangian of the n dof system will therefore have the form

$$L(\mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}} - V(\mathbf{q})$$

- The generalised force for q_i is potentially complicated because of $\mathbf{q}$ dependence in $\mathbf{M}$.

$$F_i = \frac{\partial L}{\partial q_i} = \sum_{j,k} \frac{1}{2} \frac{\partial M_{jk}}{\partial q_i} \dot{q}_j \dot{q}_k - \frac{\partial V}{\partial q_i}$$

- For an eqn. pt at $\mathbf{q} = \mathbf{q}_0$, we req. all velocities to be zero $\dot{\mathbf{q}} = 0$ and the generalised forces to vanish.

For the generalised forces to vanish, we would req.

$$\frac{\partial V}{\partial q_i} = 0$$

for each coordinate $\rightarrow$ we need a stationary pt. in the potential energy.

- consider giving the system a small ext. $\mathbf{q} = \mathbf{q}_0 + \delta \mathbf{q}$ ($\dot{\mathbf{q}} = \delta \dot{\mathbf{q}}$, $\ddot{\mathbf{q}} = \delta \ddot{\mathbf{q}}$), the Lagrangian becomes

$$L(\delta \mathbf{q}, \delta \dot{\mathbf{q}}) = \frac{1}{2} \delta \dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q}_0 + \delta \mathbf{q}) \delta \dot{\mathbf{q}} - V(\mathbf{q}_0 + \delta \mathbf{q})$$

$$= \frac{1}{2} \delta \dot{\mathbf{q}}^T \mathbf{M} \delta \dot{\mathbf{q}} - V(\mathbf{q}_0) - \frac{1}{2} \delta \mathbf{q}^T \mathbf{K} \delta \mathbf{q} + \dots$$

note terms up to second order

where $\mathbf{M}$ and $\mathbf{K}$ are constant symmetric matrices given by

$$M_{ij} = \left. \frac{\partial^2 T}{\partial \dot{q}_i \partial \dot{q}_j} \right|_{\mathbf{q}=\mathbf{q}_0} = M_{ij}(\mathbf{q}_0)$$

$$K_{ij} = \left. \frac{\partial^2 V}{\partial q_i \partial q_j} \right|_{\mathbf{q}=\mathbf{q}_0}$$

- & no linear $\delta \mathbf{q}$ terms since T itself is quadratic in $\dot{q}_j$; no linear $\delta \mathbf{q}$ terms as we are expanding about eqn. pt $\mathbf{q}_0$.

- starting the full set of generalised momenta and forces in $\mathbf{p}, \mathbf{F}$, we get.

$$\mathbf{p} = \frac{\partial L}{\partial \dot{\mathbf{q}}} = \mathbf{M} \delta \dot{\mathbf{q}}$$

$$\mathbf{F} = \frac{\partial L}{\partial \delta \mathbf{q}} = -\mathbf{K} \delta \mathbf{q}$$

and the full set of eqn. of motion can be written compactly as.

$$\mathbf{M} \delta \ddot{\mathbf{q}} = -\mathbf{K} \delta \mathbf{q}$$

- To search for normal mode solns, we try all the solns where all the coords vibrate at ω .

$$\delta \mathbf{q} = \mathbf{A} e^{i\omega t}$$

where $\mathbf{A} = (A_1, A_2, \dots, A_n)$ is a const. vector giving the shape of the mode.

- Substituting the double trial soln into the eqn. of motion, we get

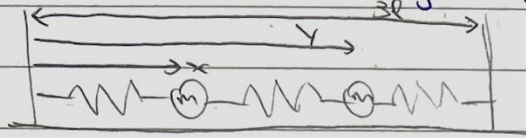
$$-\omega^2 \mathbf{M} \mathbf{A} = -\mathbf{K} \mathbf{A}$$

i.e. a generalised eigenvalue problem for $\mathbf{A}$ and ω .refer to 2A notes in details for solving $\mathbf{A}, \omega$.

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Example - spring and mass,

- consider the 2 dof system of two masses m and three springs w/ length l and spring const. k .



The Lagrangian for the system is

$$L(x, y, \dot{x}, \dot{y}) = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \dot{y}^2 - \frac{1}{2} k (x-l)^2 - \frac{1}{2} k (y-x-l)^2 - \frac{1}{2} k (3l-y-l)^2$$

Around the eqm. position,

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} l \\ 2l \end{bmatrix} + \begin{bmatrix} \delta x \\ \delta y \end{bmatrix}$$

Substituting into the Lagrangian, we get

$$\begin{aligned} L(\delta x, \delta y, \delta \dot{x}, \delta \dot{y}) &= \frac{1}{2} m \delta \dot{x}^2 + \frac{1}{2} m \delta \dot{y}^2 - \frac{1}{2} k \delta x^2 - \frac{1}{2} k (\delta x - \delta y)^2 - \frac{1}{2} k \delta y^2 \\ &= \frac{1}{2} (m \delta \dot{x}^2 + m \delta \dot{y}^2) - \frac{1}{2} (2k \delta x^2 - 2k \delta x \delta y + 2k \delta y^2) \\ &= \frac{1}{2} [\delta \dot{x} \ \delta \dot{y}] \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \delta \dot{x} \\ \delta \dot{y} \end{bmatrix} - \frac{1}{2} [\delta x \ \delta y] \begin{bmatrix} 2k & -k \\ -k & 2k \end{bmatrix} \begin{bmatrix} \delta x \\ \delta y \end{bmatrix} \end{aligned}$$

We can read off the $\underline{M}$ and $\underline{K}$ matrices as

$$\underline{M} = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \quad \underline{K} = \begin{bmatrix} 2k & -k \\ -k & 2k \end{bmatrix}$$

We want to find two normal modes in the form $\begin{bmatrix} \delta x \\ \delta y \end{bmatrix} = \underline{A} e^{i\omega t}$, so we solve

$$-\omega^2 \underline{M} \underline{A} = -\underline{K} \underline{A} \rightarrow \omega^2 \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} A_x \\ A_y \end{bmatrix} = \begin{bmatrix} 2k & -k \\ -k & 2k \end{bmatrix} \begin{bmatrix} A_x \\ A_y \end{bmatrix}$$

Method 1 By inspection, solve that $\underline{A}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\underline{A}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

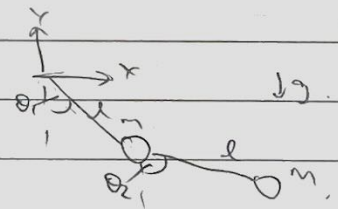
Substitute $\underline{A}_i$ into the above eqn. to solve for ω_i^2 , $\omega_1^2 = \frac{k}{m}$, $\omega_2^2 = \frac{3k}{m}$.

Method 2 solve $\det(\underline{K} - \omega^2 \underline{M}) = 0 \rightarrow (2k - \omega^2 m)^2 - k^2 = 0 \rightarrow \omega_1^2 = \frac{k}{m}$, $\omega_2^2 = \frac{3k}{m}$.

Substitute ω_i^2 into the above eqn to solve for $\underline{A}_i$, $\underline{A}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\underline{A}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Example - double pendulum,

- consider the 2 dof system of the double pendulum.



The Lagrangian for the system is

$$L(\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2) = \frac{1}{2} m l^2 (\dot{\theta}_1^2 + \dot{\theta}_2^2 + 2\dot{\theta}_1 \dot{\theta}_2 \cos(\theta_2 - \theta_1)) + m g l (2 \cos \theta_1 + \cos \theta_2)$$

Around the eqm. position,

$$\begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \delta \theta_1 \\ \delta \theta_2 \end{bmatrix}$$

Substituting into the Lagrangian, we get

$$\begin{aligned} L(\delta \theta_1, \delta \theta_2, \delta \dot{\theta}_1, \delta \dot{\theta}_2) &= \frac{1}{2} m l^2 (2 \delta \dot{\theta}_1^2 + \delta \dot{\theta}_2^2 + 2 \delta \dot{\theta}_1 \delta \dot{\theta}_2 \cos(\delta \theta_2 - \delta \theta_1)) + m g l (2 \cos \delta \theta_1 + \cos \delta \theta_2) \quad \text{const.} \\ &= \frac{1}{2} (2 m l^2 \delta \dot{\theta}_1^2 + m l^2 \delta \dot{\theta}_2^2 + 2 m l^2 \delta \dot{\theta}_1 \delta \dot{\theta}_2) - \frac{1}{2} (2 m g l \delta \theta_1^2 + m g l \delta \theta_2^2) + k \\ &= \frac{1}{2} [\delta \dot{\theta}_1 \ \delta \dot{\theta}_2] \begin{bmatrix} 2 m l^2 & m l^2 \\ m l^2 & m l^2 \end{bmatrix} \begin{bmatrix} \delta \dot{\theta}_1 \\ \delta \dot{\theta}_2 \end{bmatrix} - \frac{1}{2} [\delta \theta_1 \ \delta \theta_2] \begin{bmatrix} 2 m g l & 0 \\ 0 & m g l \end{bmatrix} \begin{bmatrix} \delta \theta_1 \\ \delta \theta_2 \end{bmatrix} + k \end{aligned}$$

We can read off the $\underline{M}$ and $\underline{K}$ matrices as

$$\underline{M} = \begin{bmatrix} 2 m l^2 & m l^2 \\ m l^2 & m l^2 \end{bmatrix} \quad \underline{K} = \begin{bmatrix} 2 m g l & 0 \\ 0 & m g l \end{bmatrix}$$

- we want to find two normal modes in the form $\begin{bmatrix} \delta x_1 \\ \delta x_2 \end{bmatrix} = A e^{i\omega t}$, so we solve.

$$-\omega^2 M A = -K A \rightarrow \omega^2 \begin{bmatrix} 2ml^2 & ml^2 \\ ml^2 & ml^2 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} 2mg & 0 \\ 0 & mg \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 2mg - \omega^2 2ml^2 & -\omega^2 ml^2 \\ -\omega^2 ml^2 & mg - \omega^2 ml^2 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = 0$$

$$\text{so } \begin{vmatrix} 2mg - \omega^2 2ml^2 & -\omega^2 ml^2 \\ -\omega^2 ml^2 & mg - \omega^2 ml^2 \end{vmatrix} = 0 \rightarrow (2g - 2\omega^2 l)(g - \omega^2 l) - \omega^4 = 0$$

$$\therefore \omega_1^2 = \frac{g}{l}(2 + \sqrt{2}) \quad \omega_2^2 = \frac{g}{l}(2 - \sqrt{2})$$

substituting into the eqn. of motion, we have.

$$\begin{bmatrix} 2mg - \frac{g}{l}(2 + \sqrt{2}) 2ml^2 & -\frac{g}{l}(2 + \sqrt{2}) ml^2 \\ -\frac{g}{l}(2 + \sqrt{2}) ml^2 & mg - \frac{g}{l}(2 + \sqrt{2}) ml^2 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} -1 + \sqrt{2} \\ 2 + \sqrt{2} \end{bmatrix}$$

$$\therefore \text{the soln is } \begin{bmatrix} \delta x_1 \\ \delta x_2 \end{bmatrix} = \begin{bmatrix} -1 + \sqrt{2} \\ 2 + \sqrt{2} \end{bmatrix} \cos\left(\sqrt{\frac{g}{l}(2 + \sqrt{2})} t + \phi_1\right) + \begin{bmatrix} -1 + \sqrt{2} \\ 2 - \sqrt{2} \end{bmatrix} \cos\left(\sqrt{\frac{g}{l}(2 - \sqrt{2})} t + \phi_2\right)$$

Zero frequency modes.

- Sometimes, one of the normal modes has $\omega^2 = 0$. In such case, the correct soln is not oscillatory, but rather has a term

$$\delta q = (C_1 + C_2 t) A$$

(note that $\delta \dot{q} = 0$).

- such a mode is a translational mode, and usually involves the entire system moving through space at a const. velocity.

Example - two masses connected by a spring.

- Consider two masses m_1, m_2 connected by a spring w/ spring constant k .

The Lagrangian of the system is.

$$L(x_1, x_2, \dot{x}_1, \dot{x}_2) = \frac{1}{2} m (\dot{x}_1^2 + \dot{x}_2^2) - \frac{1}{2} k (x_2 - x_1 - l)^2$$

Around the eqn position,

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ l \end{bmatrix} + \begin{bmatrix} \delta x_1 \\ \delta x_2 \end{bmatrix}$$

substituting into the Lagrangian, we get.

$$\begin{aligned} L(\delta x_1, \delta x_2, \delta \dot{x}_1, \delta \dot{x}_2) &= \frac{1}{2} m (\delta \dot{x}_1^2 + \delta \dot{x}_2^2) - \frac{1}{2} k (\delta x_2 - \delta x_1)^2 \\ &= \frac{1}{2} (m \delta \dot{x}_1^2 + m \delta \dot{x}_2^2) - \frac{1}{2} (k \delta x_1^2 - 2k \delta x_1 \delta x_2 + k \delta x_2^2) \\ &= \frac{1}{2} [\delta \dot{x}_1, \delta \dot{x}_2] \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \delta \dot{x}_1 \\ \delta \dot{x}_2 \end{bmatrix} - \frac{1}{2} [\delta x_1, \delta x_2] \begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \begin{bmatrix} \delta x_1 \\ \delta x_2 \end{bmatrix} \end{aligned}$$

We can read off the M and K matrices as.

$$M = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \quad K = \begin{bmatrix} k & -k \\ -k & k \end{bmatrix}$$

- We want to find two normal modes in the form $\begin{bmatrix} \delta x_1 \\ \delta x_2 \end{bmatrix} = A e^{i\omega t}$, so we solve

$$-\omega^2 M A = -K A \rightarrow \omega^2 \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$$

By inspection we have $A_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\omega^2 = 0$ and $A_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\omega^2 = \frac{2k}{m}$.

$$\therefore \text{the soln is } \begin{bmatrix} \delta x_1 \\ \delta x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} (C_1 + C_2 t) + \begin{bmatrix} -1 \\ 1 \end{bmatrix} \cos\left(\sqrt{\frac{2k}{m}} t + \phi\right)$$

zero freq. mode.

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Orthogonality of modes.

- When the $\underline{M}$ matrix is a multiple of the identity matrix, $\underline{M} = m\underline{I}$. The generalised eigenvalue problem becomes,

$$\omega^2 \underline{M} \underline{A} = \underline{K} \underline{A} \quad \rightarrow \quad \underline{K} \underline{A} = m\omega^2 \underline{A}$$

so $\underline{A}$ is a real eigenvector of $\underline{K}$.

- since $\underline{K}$ is symmetric, its eigenvectors $\underline{A}_i$ are orthogonal.
- When the $\underline{K}$ matrix is a multiple of the identity matrix, $\underline{K} = k\underline{I}$, the generalised eigenvalue problem becomes,

$$\omega^2 \underline{M} \underline{A} = \underline{K} \underline{A} \quad \rightarrow \quad \underline{M} \underline{A} = \frac{k}{\omega^2} \underline{A}$$

so $\underline{A}$ is a real eigenvector of $\underline{M}$.

- since $\underline{M}$ is symmetric, its eigenvectors $\underline{A}_i$ are orthogonal.

→ realising the modes are orthogonal can allow us to sort modes by looking for modes orthogonal to the existing solutions.

Unstable equilibria.

- In general, there is no guarantee that the eqm. we are expanding on will be a min. in the potential energy. If it is a max, we find that the ω^2 values are $-ve$, implying that $\omega = \pm \frac{i}{\tau}$ is imaginary.
- As in the one dof case, the solns change from being vibratory to being exponentially growing and decaying w/ time constant τ , i.e.

$$A e^{i\omega t} \rightarrow A e^{\pm t/\tau}$$

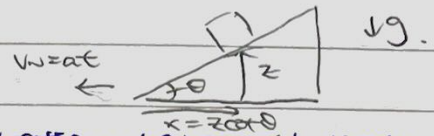
This corresponds to the eqm. solns being unstable (i.e. given a small perturbation, the perturbation will grow exponentially over the time scale τ).

- It is useful to calculate τ to get a sense of how long you have before a small perturbation becomes a very big one.

* In general, you will be at a saddle pt in potential energy, where some ω will be real, and others imaginary. → still unstable as any small perturbation involving the unstable mode will then grow exponentially.

Time-varying Lagrangians

Example - mass on an accelerating wedge



- consider a mass sliding on a wedge, which itself is accelerating in the horizontal direction at velocity $v_w = at$. What value of a is req. for the mass to remain stationary rel. to the wedge.
- The mass has one dof, z , and in cartesian coords, rel. to the wedge tip, position and velocity

$$\begin{bmatrix} x \\ z \end{bmatrix} = \begin{bmatrix} z \cot \theta \\ z \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \dot{z} \cot \theta \\ \dot{z} \end{bmatrix}$$

Since the wedge is moving horizontally at $v_w = at$, so the mental horizontal velocity of the mass is $\dot{z} \cot \theta - at$. Therefore, the Lagrangian is

$$L(z, \dot{z}, t) = \frac{1}{2} m (\dot{z}^2 + (\dot{z} \cot \theta - at)^2) - mgz$$

The generalised momentum and force are.

$$p_z = \frac{\partial L}{\partial \dot{z}} = m\dot{z} + m(\dot{z} \cot \theta - at) \cot \theta$$

$$F_z = \frac{\partial L}{\partial z} = -mg.$$

$$\therefore \frac{d}{dt}(m\dot{z} + m(\dot{z} \cot \theta - at) \cot \theta) = -mg \rightarrow \frac{\ddot{z}}{\sin^2 \theta} - \frac{a}{\tan \theta} = -g.$$

Setting $\ddot{z} = 0$, we find that we req. $a = g \tan \theta$ for the mass to remain stationary rel. to the wedge.

Forced oscillations.

- Consider a spring-mass system. The Lagrangian is

$$L(x, \dot{x}) = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} k x^2$$

and the eqn. of motion is

$$m\ddot{x} = -kx.$$

- If the system is subject to a time-varying ext. driving force $f(t)$, the Lagrangian becomes

$$L(x, \dot{x}, t) = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} k x^2 + x f(t).$$

and the eqn. of motion is

$$m\ddot{x} = -kx + f(t)$$

- The work done by the ext. force is $x f(t)$, and therefore its associated potential energy is $V = -x f(t)$.

- If we have sinusoidal applied force, $f(t) = y \cos(\omega t)$, the eqn. of motion becomes.

$$m\ddot{x} = -kx + a_0 \cos(\omega t) \rightarrow m\ddot{x} + kx = a_0 \cos(\omega t)$$

CF :

$$x_{CF} = A \cos(\sqrt{\frac{k}{m}} t + \phi)$$

PI: Let $x(t) = X e^{i\omega t}$, $f(t) = Y e^{i\omega t}$, then the eqn. of motion becomes.

$$-m\omega^2 X e^{i\omega t} + k X e^{i\omega t} = Y e^{i\omega t} \rightarrow \frac{Y}{X} = \frac{1/m}{\frac{k}{m} - \omega^2}$$

$$\therefore x_{PI} = a_0 \frac{Y}{X} \cos(\omega t) = \frac{a_0/m}{\frac{k}{m} - \omega^2} \cos(\omega t).$$

$$\rightarrow x = x_{CF} + x_{PI} = A \cos(\sqrt{\frac{k}{m}} t + \phi) + \frac{a_0/m}{\frac{k}{m} - \omega^2} \cos(\omega t)$$

- x_{PI} diverges when $\omega = \sqrt{\frac{k}{m}}$, i.e. frequency of driving frequency = natural frequency $\rightarrow$ resonance.

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Parametric resonance.

- consider a system w/ an eqn. of motion of the form

$$\ddot{q} = -\omega^2(t)q.$$

We will only focus on systems where $\omega^2(t) = \omega_0^2(1 + h\cos(\omega t))$, where $h \ll 1$ is small, i.e.

$$\ddot{q} = -\omega_0^2(1 + h\cos(\omega t))q.$$

This eqn. of motion is Mathieu's eqn. and does not admit simple closed solns.

- we can rewrite the eqn. of motion for the system in the form.

$$\ddot{q} + \omega_0^2 q = h\omega_0^2 \cos(\omega t)q.$$

which is the standard form for a driven oscillator, but w/ the driving force $f(t) = h\omega_0^2 \cos(\omega t)q$.

- if the system is oscillating at its natural frequency, $q = A\cos(\omega_0 t)$, the driving force looks like.

$$f(t) = h\omega_0^2 \cos(\omega t) A \cos(\omega_0 t) = \frac{hA}{2} \cos((\omega - \omega_0)t) + \cos((\omega + \omega_0)t)$$

- if we have $\omega = 2\omega_0$, then one of the terms looks like a driving force of resonant frequency $\omega - \omega_0 = \omega_0 \rightarrow$ resonance! (other term is not important).

• Force is prop. to amplitude, so the amplitude increases exponentially.

- if we have $\omega = \omega_0$, then neither of the terms look like a driving force of resonance $\rightarrow$ no resonance.

Example - mass-spring system

- consider a mass-spring system, where the spring constant is a function of time $k(t)$.

The Lagrangian for the system is

$$L(x, \dot{x}, t) = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}k(t)x^2$$

The generalized momentum and force are

$$p_x = \frac{\partial L}{\partial \dot{x}} = m\dot{x}$$

$$F_x = \frac{\partial L}{\partial x} = -k(t)x$$

$$\therefore \frac{d}{dt}(m\dot{x}) = -k(t)x \rightarrow \ddot{x} = -\frac{k(t)}{m}x$$

which is of the form

$$\ddot{q} = -\omega^2(t)q.$$

Using a computer

Left and right Riemann sum

- Suppose we know the function $v(t)$ and wish to calculate:

$$x = \int_{t_0}^{t_1} v(t) dt$$

The integral is equivalent to the area under the curve

- To calculate the integral, we break it into N small time intervals, each of duration $\Delta t = \frac{t_1 - t_0}{N}$.

Defining the set of boundary times $t_{n+1} = t_n + \Delta t$, the integral becomes a sum of little integrals.

$$x = \int_{t_0}^{t_1} v(t) dt = \sum_{n=0}^{N-1} \int_{t_n}^{t_{n+1}} v(t) dt$$

- The simplest approximation replaces each area w/ a rectangle. The height of the rectangle as the height of the function at the LHS / RHS.

Left Riemann sum

$$\int_{t_n}^{t_{n+1}} v(t) dt \approx v(t_n) \Delta t$$

Right Riemann sum

$$\int_{t_n}^{t_{n+1}} v(t) dt \approx v(t_{n+1}) \Delta t$$

We can estimate the error by Taylor expanding around t_n / t_{n+1} .

$$v(t) = v(t_n) + (t - t_n) v'(t_n) + \dots$$

$$v(t) = v(t_{n+1}) + (t - t_{n+1}) v'(t_{n+1})$$

so the integral becomes:

$$\begin{aligned} \int_{t_n}^{t_{n+1}} v(t) dt &= \int_{t_n}^{t_{n+1}} [v(t_n) + (t - t_n) v'(t_n) + \dots] dt \\ &= v(t_n) \Delta t + \frac{1}{2} v'(t_n) \Delta t^2 + \dots \end{aligned}$$

$$\begin{aligned} \int_{t_n}^{t_{n+1}} v(t) dt &= \int_{t_n}^{t_{n+1}} [v(t_{n+1}) + (t - t_{n+1}) v'(t_{n+1}) + \dots] dt \\ &= v(t_{n+1}) \Delta t - \frac{1}{2} v'(t_{n+1}) \Delta t^2 + \dots \end{aligned}$$

This approximation throws away a term of size $O(\Delta t^2)$ in each rectangle. However, the whole integral contains $N = \frac{t_1 - t_0}{\Delta t}$ rectangles, so the total error will be of size $N \Delta t^2 \sim O(\Delta t)$. $\rightarrow$ first order method.

Mid-pt. method and Simpson's method.

- The mid-pt. method uses the height of the mid-pt as the height of the rectangle.

$$\int_{t_n}^{t_{n+1}} v(t) dt \approx v(t_n + \frac{\Delta t}{2}) \Delta t$$

We can estimate the error by Taylor expanding around $t_n + \frac{\Delta t}{2}$

$$\int_{t_n}^{t_{n+1}} v(t) dt = \int_{t_n}^{t_{n+1}} [v(t_n + \frac{\Delta t}{2}) + \underbrace{v'(t_n + \frac{\Delta t}{2})}_{\text{integrates to zero}} (t - (t_n + \frac{\Delta t}{2})) + \dots] dt = v(t_n + \frac{\Delta t}{2}) \Delta t + O(\Delta t^3)$$

so the error on each rectangle is $O(\Delta t^3)$, and the total error is $N O(\Delta t^3) \sim O(\Delta t^2)$. $\rightarrow$ second order method.

- Simpson's rule is a higher order method

$$\int_{t_n}^{t_{n+1}} v(t) dt \approx \frac{1}{6} (v(t_n) + 4v((t_n + t_{n+1})/2) + v(t_{n+1})) \Delta t$$

where the error on each rectangle is $O(\Delta t^5)$, and the total error is $N O(\Delta t^5) \sim O(\Delta t^4)$. $\rightarrow$ fourth order method.

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Forward and backward (implicit) Euler.

- In mechanics, we need to integrate a DE for the motion, rather than computing a standard area integral.
e.g. we want to find $y(t)$ that satisfies the first order ODE

$$\frac{dy}{dt} = f(y, t)$$

(If $f(y, t) = f(t)$ doesn't depend on y , then it would be a standard area integral).

- The forward Euler method is similar to the left Riemann sum, but allows for y dependence in $f(y, t)$.

$$y(t_{n+1}) = y(t_n) + f(y(t_n), t_n) \Delta t$$

We can estimate the error by Taylor expanding around t_n

$$y(t_{n+1}) = y(t_n) + f(y(t_n), t_n) \Delta t + O(\Delta t^2)$$

so the error of each time step is $O(\Delta t^2)$, and the total error is $N O(\Delta t^2) \sim O(\Delta t)$ → first order method.

- In mechanics applications, we have $\dot{x} = v(t)$, $\dot{v} = \frac{F(x, v, t)}{m} = a(x, v, t)$

$$x(t_{n+1}) = x(t_n) + v(t_n) \Delta t$$

$$v(t_{n+1}) = v(t_n) + a(x(t_n), v(t_n), t_n) \Delta t$$

- For Forward Euler method.

↳ ✓ intuitive, easy to code

↳ ✗ $O(\Delta t)$ method → req. v. small Δt to be accurate

↳ ✗ small error in position → end up in wrong position → wrong force.

- The backward Euler method is similar to the right Riemann sum, but allows for y dependence in $f(y, t)$.

$$y(t_{n+1}) = y(t_n) + f(y(t_{n+1}), t_{n+1}) \Delta t$$

We can estimate the error by Taylor expanding around t_n

$$y(t_{n+1}) = y(t_n) + f(y(t_{n+1}), t_{n+1}) \Delta t + O(\Delta t^2)$$

so the error of each time step is $O(\Delta t^2)$, and the total error is $N O(\Delta t^2) \sim O(\Delta t)$ → first order method

- In mechanics applications, we have $\dot{x} = v(x)$, $\dot{v} = \frac{F(x, v, t)}{m} = a(x, v, t)$

$$x(t_{n+1}) = x(t_n) + v(t_{n+1}) \Delta t$$

$$v(t_{n+1}) = v(t_n) + a(x(t_{n+1}), v(t_{n+1}), t_{n+1}) \Delta t$$

- For Backward Euler method

↳ ✓ more numerically stable than forward Euler

↳ ✗ $O(\Delta t)$ method → req. v. small Δt to be accurate

↳ ✗ more work for the programmer and the computer.

Leapfrog.

- The leapfrog method is restricted to dynamics problems where the forces depend only on position, i.e. $\vec{x} = \vec{x}(t)$, $\vec{v} = \frac{\vec{F}(\vec{x}, t)}{m} = \vec{a}(\vec{x}, t)$.
- The leapfrog method works by interleaving our time pos t_n w/ midpt times $t_{n+1/2} = t_n + \frac{\Delta t}{2}$. We then evaluate velocities at the mid pts, and positions at the integer pts.

$$\vec{x}(t_{n+1}) = \vec{x}(t_n) + \vec{v}(t_{n+1/2}) \Delta t$$

$$\vec{v}(t_{n+1/2}) = \vec{v}(t_{n-1/2}) + \vec{a}(\vec{x}(t_n), t_n) \Delta t$$

- By doing this, the position and velocity are both updated by a whole time step based on their rate of change at the mid pt of their time increment.
- We can estimate the error by Taylor expanding around $t_n + \frac{\Delta t}{2}$.

$$\begin{aligned} \vec{x}(t_{n+1}) &= \vec{x}(t_n) = \int_{t_n}^{t_{n+1}} \vec{v} dt \\ &= \vec{x}(t_n) + \int_{t_n}^{t_{n+1}} \vec{v}(t_{n+1/2}) + \underbrace{\vec{v}'(t_{n+1/2})(t - t_{n+1/2})}_{\text{integrates to zero}} + \frac{1}{2} \vec{v}''(t_{n+1/2})(t - t_{n+1/2})^2 + \dots dt \\ &= \vec{x}(t_n) + \vec{v}(t_{n+1/2}) \Delta t + O(\Delta t^3) \end{aligned}$$

$$\begin{aligned} \vec{v}(t_{n+1/2}) &= \vec{v}(t_{n-1/2}) + \int_{t_{n-1/2}}^{t_{n+1/2}} \vec{a} dt \\ &= \vec{v}(t_{n-1/2}) + \int_{t_{n-1/2}}^{t_{n+1/2}} \vec{a}(t_n) + \underbrace{\vec{a}'(t_n)(t - t_n)}_{\text{integrates to zero}} + \frac{1}{2} \vec{a}''(t_n)(t - t_n)^2 + \dots dt \\ &= \vec{v}(t_{n-1/2}) + \vec{a}(t_n) \Delta t + O(\Delta t^3) \end{aligned}$$

so the error of each step is $O(\Delta t^3)$, and the total error is $N O(\Delta t^3) \sim O(\Delta t^2) \rightarrow$ second order method.